

GRADE 7 ACCELERATED · UNIT 3

Exponents and Scientific Notation

Ten lessons in teaching order · the next slide jumps to each one

Contents

3.01	Product Laws of Exponents	slide 3
3.02	Quotient Laws of Exponents	slide 38
3.03	Exponential Expressions	slide 72
3.04	Negative Exponent Law	slide 106
3.05	Applying Exponent Laws	slide 140
3.06–07	Evaluating and Equivalent Expressions	slide 174
T-A1	Exponent Laws with Variable Bases	slide 209
T-A2	Negative Exponents with Variable Bases	slide 243
3.08	Writing Large Numbers in Scientific Notation	slide 277
3.09	Writing Small Numbers in Scientific Notation	slide 311

Product Laws of Exponents

MA.7.NSO.1.1

I can use the product of powers, power of a product, power of a power and identity laws to rewrite and evaluate a numerical expression.

Warm-Up

1. $\sqrt{81}$
2. Between which two whole numbers is $\sqrt{40}$?
3. Solve $3x - 7 = 11$
4. Evaluate $\left(\frac{2}{3}\right)^2$

Warm-Up

1. $\sqrt{81}$ **9**

2. Between which two whole numbers is $\sqrt{40}$? **6 and 7**

3. Solve $3x - 7 = 11$ **$x = 6$**

4. Evaluate $\left(\frac{2}{3}\right)^2$ **$\frac{4}{9}$**

Notes

I. A power is a shorthand for repeated multiplication

- A. **base** — the number used as a factor
- B. **exponent** — how many times the base is a factor
- C. **power** — the whole thing, base and exponent together

$$2^3 = 2 \cdot 2 \cdot 2 = 8 \qquad \left(\frac{2}{3}\right)^2 = \frac{2}{3} \cdot \frac{2}{3} = \frac{4}{9}$$

Notes

II. Product of powers — same base, add the exponents

$$3^2 \cdot 3^3 = (3 \cdot 3)(3 \cdot 3 \cdot 3) = 3^5$$

$$3^2 \cdot 2^3 = (3 \cdot 3)(2 \cdot 2 \cdot 2) \text{ — different bases, no shortcut}$$

Product of powers: $a^m \cdot a^n = a^{m+n}$

Check with values: $3^2 \cdot 3^3 = 9 \cdot 27 = 243 = 3^5$

Notes

III. Power of a product — the exponent goes to every factor

$$(2 \cdot 5)^3 = (2 \cdot 5)(2 \cdot 5)(2 \cdot 5) = (2 \cdot 2 \cdot 2)(5 \cdot 5 \cdot 5) = 2^3 \cdot 5^3$$

Power of a product: $(ab)^m = a^m \cdot b^m$

Works for a product only, never for a sum:

$$(10 + (-2))^3 = 8^3 = 512 \quad \text{but} \quad 10^3 + (-2)^3 = 992$$

Notes

IV. Power of a power, and the exponent 1

Power of a power	Repeated multiplication	Single power
$(4^2)^3$	$4^2 \cdot 4^2 \cdot 4^2$	4^6
$(10^4)^4$	$10^4 \cdot 10^4 \cdot 10^4 \cdot 10^4$	10^{16}
$(3^1)^5$	$3 \cdot 3 \cdot 3 \cdot 3 \cdot 3$	3^5

Power of a power: $(a^m)^n = a^{mn}$ Identity: $a^1 = a$

Example 1

$$3^2 \cdot 3^3$$

Which law? Then find the value.

Example 1

$$3^2 \cdot 3^3 = 3^{2+3} = 3^5$$

$$3^5 = 243$$

$$9 \cdot 27 = 243$$

Answer: 243

Your Turn

$$2^3 \cdot 2^4$$

Which law? Then find the value.

Your Turn

$$2^3 \cdot 2^4$$

Which law? Then find the value.

$$2^3 \cdot 2^4 = 2^{3+4} = 2^7$$

$$2^7 = 128$$

Answer: $2^7 = 128$

Example 2

$$[(-2)^3]^2$$

Which law? Then find the value.

Example 2

$$[(-2)^3]^2 = (-2)^{3 \cdot 2} = (-2)^6$$

$$(-2)^6 = 64$$

$$(-8)^2 = 64$$

Answer: 64

Your Turn

$$[(-3)^2]^3$$

Which law? Then find the value.

Your Turn

$$[(-3)^2]^3$$

Which law? Then find the value.

$$((-3)^2)^3 = (-3)^{2 \cdot 3} = (-3)^6$$

$$(-3)^6 = 729$$

Answer: $(-3)^6 = 729$

Whiteboards · Question 1 of 9

$$5^2 \cdot 5^3$$

Answer it.

Whiteboards · Question 1 of 9

$$5^2 \cdot 5^3$$

$$5^2 \cdot 5^3 = 5^{2+3} = 5^5$$

Answer: 5^5

Whiteboards • Question 2 of 9

A rectangle is 2^4 centimeters long and 2^2 centimeters wide.

Its perimeter is 40 centimeters.

What is the area of the rectangle, in square centimeters?

Answer it.

Whiteboards • Question 2 of 9

A rectangle is 2^4 centimeters long and 2^2 centimeters wide.

Its perimeter is 40 centimeters.

What is the area of the rectangle, in square centimeters?

Area = length $\times$ width

$$2^4 \cdot 2^2 = 2^{4+2} = 2^6$$

$$2^6 = 64$$

Answer: $2^6 = 64$ square centimeters

Whiteboards · Question 3 of 9

$$(2 \cdot 5)^3$$

Answer it.

Whiteboards · Question 3 of 9

$$(2 \cdot 5)^3$$

$$(2 \cdot 5)^3 = 2^3 \cdot 5^3 = 8 \cdot 125 = 1,000$$

Answer: 1000

Whiteboards · Question 4 of 9

$$(3^2)^4$$

Answer it.

Whiteboards • Question 4 of 9

$$(3^2)^4$$

$$(3^2)^4 = 3^{2 \cdot 4} = 3^8$$

Answer: 3^8

Whiteboards • Question 5 of 9

$$4^3 \cdot 4^2$$

Which expression is equivalent?

A. 16^5

B. 8^5

C. 4^6

D. 4^5

Answer it.

Whiteboards · Question 5 of 9

$$4^3 \cdot 4^2$$

Which expression is equivalent?

$$4^3 \cdot 4^2 = 4^{3+2} = 4^5$$

Answer: D — 4^5

Whiteboards • Question 6 of 9

$$(2^3)^2$$

Which expression is equivalent?

A. 2^5

B. 4^6

C. 2^6

D. 2^9

Answer it.

Whiteboards • Question 6 of 9

$$(2^3)^2$$

Which expression is equivalent?

$$(2^3)^2 = 2^{3 \cdot 2} = 2^6$$

Answer: C — 2^6

Whiteboards · Question 7 of 9

$$(-3)^2 \cdot (-3)^1$$

Answer it.

Whiteboards • Question 7 of 9

$$(-3)^2 \cdot (-3)^1$$

$$(-3)^2 \cdot (-3)^1 = (-3)^{2+1} = (-3)^3$$

$$(-3)^3 = -27$$

Answer: -27

Whiteboards • Question 8 of 9

$$\left(\frac{1}{2}\right)^2 \cdot \left(\frac{1}{2}\right)^3$$

Answer it.

Whiteboards • Question 8 of 9

$$\left(\frac{1}{2}\right)^2 \cdot \left(\frac{1}{2}\right)^3$$

$$\left(\frac{1}{2}\right)^2 \cdot \left(\frac{1}{2}\right)^3 = \left(\frac{1}{2}\right)^{2+3} = \left(\frac{1}{2}\right)^5$$

$$\left(\frac{1}{2}\right)^5 = \frac{1}{32}$$

Answer: $\frac{1}{32}$

Whiteboards • Question 9 of 9

Aiyanna says she can use the power of a product law to evaluate $(10 + (-2))^3$. Is she right?

Write your answer in sentences.

Whiteboards • Question 9 of 9

Aiyanna says she can use the power of a product law to evaluate $(10 + (-2))^3$. Is she right?

Inside the parentheses is a sum, not a product.

$$10 + (-2) = 8$$

$$8^3 = 512$$

Answer: No — $(10 + (-2))^3 = 8^3 = 512$

Independent Set

1. Write as one power. $3^4 \cdot 3^5$
2. Find the value. $2^3 \cdot 2^2$
3. Write as one power. $(7^3)^2$
4. Find the value. $(3 \cdot 4)^2$
5. Find the value. $(-2)^3 \cdot (-2)^2$
6. Find the value of n . $5^2 \cdot 5^n = 5^7$

IXL

1. Open IXL and start today's skills.
2. Work on paper where the question needs work. The answer box does not show it.
3. Every skill listed is required, to a SmartScore of 67. Due at the start of the next class.

Today's skills — all required

Evaluate powers — SZQ

Multiplication with exponents — 5SK

Power rule — 8DA

Understanding exponents — BFA

Quotient Laws of Exponents

MA.7.NSO.1.1

I can use the quotient of powers, power of a quotient and zero exponent laws to rewrite and evaluate a numerical expression.

Warm-Up

1. Write as one power: $3^2 \cdot 3^3$
2. $\sqrt[3]{-27}$
3. Solve $2x + 3 < 11$
4. Simplify $\frac{5 \cdot 5 \cdot 5}{5 \cdot 5}$

Warm-Up

1. Write as one power: $3^2 \cdot 3^3$ 3^5
2. $\sqrt[3]{-27}$ -3
3. Solve $2x + 3 < 11$ $x < 4$
4. Simplify $\frac{5 \cdot 5 \cdot 5}{5 \cdot 5}$ 5

Notes

I. Quotient of powers — same base, subtract the exponents

$$\frac{5^6}{5^3} = \frac{5 \cdot 5 \cdot 5 \cdot 5 \cdot 5 \cdot 5}{5 \cdot 5 \cdot 5} = 5 \cdot 5 \cdot 5 = 5^3$$

Quotient of powers: $\frac{a^m}{a^n} = a^{m-n} \quad (a \neq 0)$

Check with values: $\frac{5^6}{5^3} = \frac{15625}{125} = 125 = 5^3$

Notes

II. Power of a quotient — the exponent goes top and bottom

$$\left(\frac{8}{10}\right)^3 = \frac{8}{10} \cdot \frac{8}{10} \cdot \frac{8}{10} = \frac{8 \cdot 8 \cdot 8}{10 \cdot 10 \cdot 10} = \frac{8^3}{10^3}$$

Power of a quotient: $\left(\frac{a}{b}\right)^m = \frac{a^m}{b^m} \quad (b \neq 0)$

Either way: $\frac{8^3}{10^3} = \frac{512}{1000} = 0.512$ and $(0.8)^3 = 0.512$

Notes

III. The zero exponent — two roads to the same answer

2^4	2^3	2^2	2^1	2^0
16	8	4	2	1

$$\frac{2^3}{2^3} = \frac{8}{8} = 1 \quad \text{and} \quad \frac{2^3}{2^3} = 2^{3-3} = 2^0 \quad \text{so} \quad 2^0 = 1$$

Zero exponent: $a^0 = 1$ for every nonzero a

Example 1

$$\frac{(-2)^6}{(-2)^3}$$

Which law? Then find the value.

Example 1

$$\frac{(-2)^6}{(-2)^3} = (-2)^{6-3} = (-2)^3$$

$$(-2)^3 = -8$$

$$64 \div (-8) = -8$$

Answer: -8

Your Turn

$$\frac{3^6}{3^2}$$

Which law? Then find the value.

Your Turn

$$\frac{3^6}{3^2}$$

Which law? Then find the value.

$$\frac{3^6}{3^2} = 3^{6-2} = 3^4$$

$$3^4 = 81$$

Answer: $3^4 = 81$

Example 2

$$\left(\frac{0.2}{10}\right)^4$$

Find the value.

Example 2

$$\left(\frac{0.2}{10}\right)^4 = \frac{0.2^4}{10^4} = \frac{0.0016}{10000}$$

$$\frac{0.0016}{10000} = 0.00000016$$

$$(0.02)^4 = 0.00000016$$

Answer: 0.00000016

Your Turn

$$\left(\frac{3}{4}\right)^3$$

Find the value.

Your Turn

$$\left(\frac{3}{4}\right)^3$$

Find the value.

$$\left(\frac{3}{4}\right)^3 = \frac{3^3}{4^3} = \frac{27}{64}$$

Answer: $\frac{27}{64}$

Whiteboards · Question 1 of 9

$$\frac{10^5}{10^3}$$

Answer it.

Whiteboards • Question 1 of 9

$$\frac{10^5}{10^3}$$

$$\frac{10^5}{10^3} = 10^{5-3} = 10^2$$

Answer: 10^2

Whiteboards • Question 2 of 9

A website puts 8^4 photos into 8 equal albums.

The site is 30 days old.

How many photos are in each album?

Answer it.

Whiteboards • Question 2 of 9

A website puts 8^4 photos into 8 equal albums.

The site is 30 days old.

How many photos are in each album?

$$8^4 \div 8^1 = 8^{4-1} = 8^3$$

$$8^3 = 512$$

Answer: 512

Whiteboards • Question 3 of 9

$$\frac{(-6)^6}{(-6)^5}$$

Answer it.

Whiteboards • Question 3 of 9

$$\frac{(-6)^6}{(-6)^5}$$

$$\frac{(-6)^6}{(-6)^5} = (-6)^{6-5} = (-6)^1 = -6$$

Answer: -6

Whiteboards • Question 4 of 9

$$\left(\frac{-5}{10}\right)^7$$

Answer it.

Whiteboards • Question 4 of 9

$$\left(\frac{-5}{10}\right)^7$$

$$\left(\frac{-5}{10}\right)^7 = \frac{(-5)^7}{10^7}$$

Answer: $\frac{(-5)^7}{10^7}$

Whiteboards • Question 5 of 9

$$9^0$$

Which is the value?

A. 1

B. 9

C. $1/9$

D. 0

Answer it.

Whiteboards • Question 5 of 9

$$9^0$$

Which is the value?

Any nonzero number to the power 0 is 1.

$$9^0 = 1$$

Answer: A — 1

Whiteboards • Question 6 of 9

$$\frac{4^8}{4^2}$$

Which expression is equivalent?

A. 1^6

B. 4^{10}

C. 4^6

D. 4^4

Answer it.

Whiteboards • Question 6 of 9

$$\frac{4^8}{4^2}$$

Which expression is equivalent?

$$\frac{4^8}{4^2} = 4^{8-2} = 4^6$$

Answer: C — 4^6

Whiteboards • Question 7 of 9

$$\frac{0.5^8}{0.5^5}$$

Answer it.

Whiteboards • Question 7 of 9

$$\frac{0.5^8}{0.5^5}$$

$$\frac{0.5^8}{0.5^5} = 0.5^{8-5} = 0.5^3$$

$$0.5^3 = 0.125$$

Answer: 0.125

Whiteboards • Question 8 of 9

$$\left(\frac{10}{11}\right)^0$$

Answer it.

Whiteboards • Question 8 of 9

$$\left(\frac{10}{11}\right)^0$$

Any nonzero number to the power 0 is 1.

$$\left(\frac{10}{11}\right)^0 = 1$$

Answer: 1

Whiteboards • Question 9 of 9

Hannah cancels: $\frac{2^3}{2^3} = \frac{8}{8} = 1$. Sebastian subtracts: $\frac{2^3}{2^3} = 2^0$.

Who is right?

Write your answer in sentences.

Whiteboards • Question 9 of 9

Hannah cancels: $\frac{2^3}{2^3} = \frac{8}{8} = 1$. Sebastian subtracts: $\frac{2^3}{2^3} = 2^0$.

Who is right?

Hannah: $\frac{2^3}{2^3} = \frac{8}{8} = 1$

Sebastian: $\frac{2^3}{2^3} = 2^{3-3} = 2^0$

Same fraction, so $2^0 = 1$

Answer: Both — $2^0 = 1$

Independent Set

1. Write as one power. $\frac{6^7}{6^3}$
2. Find the value. $\frac{5^6}{5^4}$
3. Find the value. $\frac{(-4)^5}{(-4)^3}$
4. Rewrite so the exponent is on the numerator and on the denominator. $\left(\frac{3}{7}\right)^4$
5. Find the value. $\left(\frac{12}{13}\right)^0$
6. Find the value of n . $\frac{2^9}{2^n} = 2^4$

IXL

1. Open IXL and start today's skills.
2. Work on paper where the question needs work. The answer box does not show it.
3. Every skill listed is required, to a SmartScore of 67. Due at the start of the next class.

Today's skills — all required

Powers with negative bases — WFW

Division with exponents — XNC

Identify equivalent expressions involving whole-number exponents — KBE

Exponential Expressions

MA.7.NSO.1.1

I can apply more than one law of exponents to evaluate an expression and to write an equivalent one.

Warm-Up

1. Find the value: $\frac{10^5}{10^3}$
2. Is $\sqrt{50}$ closer to 7 or to 8?
3. Solve $4(x - 2) = 12$
4. Evaluate $2 + 3 \cdot 4^2$

Warm-Up

1. Find the value: $\frac{10^5}{10^3}$ **100**
2. Is $\sqrt{50}$ closer to 7 or to 8? **7**
3. Solve $4(x - 2) = 12$ **$x = 5$**
4. Evaluate $2 + 3 \cdot 4^2$ **50**

Notes

I. The seven laws, in one place

Law	Rule	Example
Product of powers	$a^m \cdot a^n = a^{m+n}$	$2^3 \cdot 2^1 = 2^4$
Quotient of powers	$a^m \div a^n = a^{m-n}$	$(-3)^7 \div (-3)^2 = (-3)^5$
Power of a product	$(ab)^m = a^m \cdot b^m$	$(-7 \cdot 3)^2 = (-7)^2 \cdot 3^2$
Power of a quotient	$(a \div b)^m = a^m \div b^m$	$(13 \div 5)^3 = 13^3 \div 5^3$
Power of a power	$(a^m)^n = a^{mn}$	$(6^3)^4 = 6^{12}$
Identity exponent	$a^1 = a$	$15^1 = 15$
Zero exponent	$a^0 = 1$	$127^0 = 1$

Notes

II. More than one law — make a plan, then follow it

$$\text{Winston: } \left(\frac{2^6}{2^4} \right)^3 = (2^2)^3 = 2^6 = 64$$

$$\text{Yvette: } \left(\frac{2^6}{2^4} \right)^3 = \frac{(2^6)^3}{(2^4)^3} = \frac{2^{18}}{2^{12}} = 2^6 = 64$$

Different plans, same value. Any correct sequence of laws gives the same answer.

Notes

III. Order of operations still applies

- A. Parentheses first — including a fraction bar, which is a pair of parentheses in disguise.
- B. Exponents next — the laws of exponents are applied here.
- C. Then multiply and divide, left to right; then add and subtract.
- D. **Never across a + or a -:** $2^3 + 2^2 = 8 + 4 = 12$, not 2^5 .

Example 1

$$\frac{4^5 \cdot 4^0}{4^2}$$

Make a plan. Then find the value.

Example 1

$$\frac{4^5 \cdot 4^0}{4^2} = \frac{4^{5+0}}{4^2} = \frac{4^5}{4^2}$$

$$\frac{4^5 \cdot 4^0}{4^2} = \frac{4^5 \cdot 1}{4^2} = \frac{4^5}{4^2}$$

$$\frac{4^5}{4^2} = 4^3 = 64$$

Answer: 64

Your Turn

$$\frac{5^4 \cdot 5^0}{5^2}$$

Make a plan. Then find the value.

Your Turn

$$\frac{5^4 \cdot 5^0}{5^2}$$

Make a plan. Then find the value.

$$5^0 = 1, \text{ so the top is } 5^4 \cdot 1 = 5^4$$

$$\frac{5^4}{5^2} = 5^{4-2} = 5^2 = 25$$

$$\textbf{Answer: } 5^2 = 25$$

Example 2

$$11.6^0 \cdot \frac{(-15)^4}{1.5^4}$$

Make a plan. Then find the value.

Example 2

$$11.6^0 \cdot \frac{(-15)^4}{1.5^4} = 1 \cdot \left(\frac{-15}{1.5}\right)^4$$

$$\left(\frac{-15}{1.5}\right)^4 = (-10)^4$$

$$(-10)^4 = 10000$$

Answer: 10,000

Your Turn

$$7^0 \cdot \frac{(-12)^3}{4^3}$$

Make a plan. Then find the value.

Your Turn

$$7^0 \cdot \frac{(-12)^3}{4^3}$$

Make a plan. Then find the value.

$$7^0 = 1$$

$$\frac{(-12)^3}{4^3} = \left(\frac{-12}{4}\right)^3 = (-3)^3 = -27$$

Answer: -27

Whiteboards • Question 1 of 9

A warehouse holds 6^4 boxes in 6^2 equal rows.

Each box weighs 3 kilograms.

How many boxes are in each row?

Answer it.

Whiteboards • Question 1 of 9

A warehouse holds 6^4 boxes in 6^2 equal rows.

Each box weighs 3 kilograms.

How many boxes are in each row?

$$6^4 \div 6^2 = 6^{4-2} = 6^2$$

$$6^2 = 36$$

Answer: 36

Whiteboards · Question 2 of 9

$$(3^0 \cdot 3^2)^3$$

Answer it.

Whiteboards • Question 2 of 9

$$(3^0 \cdot 3^2)^3$$

Inside first: $3^0 \cdot 3^2 = 3^{0+2} = 3^2$

$$(3^2)^3 = 3^{2 \cdot 3} = 3^6 = 729$$

Answer: 729

Whiteboards • Question 3 of 9

$$\frac{(0.5^3)^2}{(0.5)^2}$$

Answer it.

Whiteboards • Question 3 of 9

$$\frac{(0.5^3)^2}{(0.5)^2}$$

$$\text{Top: } (0.5^3)^2 = 0.5^{3 \cdot 2} = 0.5^6$$

$$\frac{0.5^6}{0.5^2} = 0.5^{6-2} = 0.5^4 = 0.0625$$

Answer: 0.0625

Whiteboards • Question 4 of 9

$$\left(\frac{2.5^5}{2.5^3}\right)^1$$

Answer it.

Whiteboards • Question 4 of 9

$$\left(\frac{2.5^5}{2.5^3} \right)^1$$

$$\frac{2.5^5}{2.5^3} = 2.5^{5-3} = 2.5^2$$

$$(2.5^2)^1 = 2.5^2 = 6.25$$

Answer: 6.25

Whiteboards • Question 5 of 9

$$\frac{(2^2 \cdot 3 \cdot 2^0)^6}{2^{10}}$$

Which is the value?

A. 1/4

B. 2^{22}

C. 0

D. 4

Answer it.

Whiteboards • Question 5 of 9

$$\frac{(2^2 \cdot 3 \cdot 2^0)^6}{2^{10}}$$

Which is the value?

$$3 \cdot 2^0 = 1, \text{ so the top is } (2^2)^6 = 2^{2 \cdot 6} = 2^{12}$$

$$\frac{2^{12}}{2^{10}} = 2^{12-10} = 2^2 = 4$$

Answer: D — 4

Whiteboards • Question 6 of 9

$$\frac{0.2^3}{2^3} \cdot 0.1^2$$

Which expression is equivalent?

A. $0.2^0 \cdot 0.1^2$

B. $0.2^3 \cdot 0.1^2$

C. 0.1^5

D. 0.1^6

Answer it.

Whiteboards • Question 6 of 9

$$\frac{0.2^3}{2^3} \cdot 0.1^2$$

Which expression is equivalent?

$$\frac{0.2^3}{2^3} = \left(\frac{0.2}{2}\right)^3 = 0.1^3$$

$$0.1^3 \cdot 0.1^2 = 0.1^{3+2} = 0.1^5$$

Answer: C — 0.1^5

Whiteboards • Question 7 of 9

$$0.1^2 \cdot (0.1^1)^2$$

Answer it.

Whiteboards • Question 7 of 9

$$0.1^2 \cdot (0.1^1)^2$$

$$(0.1^1)^2 = 0.1^{1 \cdot 2} = 0.1^2$$

$$0.1^2 \cdot 0.1^2 = 0.1^{2+2} = 0.1^4 = 0.0001$$

Answer: 0.0001

Whiteboards • Question 8 of 9

$$\frac{(-4)^2 \cdot (-4)^1}{(-4)^0}$$

Answer it.

Whiteboards • Question 8 of 9

$$\frac{(-4)^2 \cdot (-4)^1}{(-4)^0}$$

Top: $(-4)^2 \cdot (-4)^1 = (-4)^{2+1} = (-4)^3$

Bottom: $(-4)^0 = 1$

$$(-4)^3 = -64$$

Answer: -64

Whiteboards • Question 9 of 9

Winston writes $\left(\frac{2^6}{2^4}\right)^3 = (2^2)^3$. Yvette writes $\left(\frac{2^6}{2^4}\right)^3 = \frac{2^{18}}{2^{12}}$.

Do they get the same value? Explain.

Write your answer in sentences.

Whiteboards • Question 9 of 9

Winston writes $\left(\frac{2^6}{2^4}\right)^3 = (2^2)^3$. Yvette writes $\left(\frac{2^6}{2^4}\right)^3 = \frac{2^{18}}{2^{12}}$.

Do they get the same value? Explain.

Winston: $(2^2)^3 = 2^{2 \cdot 3} = 2^6$

Yvette: $\frac{2^{18}}{2^{12}} = 2^{18-12} = 2^6$

$$2^6 = 64$$

Answer: Yes — both give $2^6 = 64$

Independent Set

1. Find the value. $\frac{4^2 \cdot 4^0}{4^1}$
2. Find the value. $(2^0 \cdot 2^3)^2$
3. Write as one power. $\left(\frac{5^6}{5^4}\right)^3$
4. Find the value. $0.2^2 \cdot (0.2^1)^2$
5. Find the value. $\frac{(-3)^3 \cdot (-3)^0}{(-3)^2}$
6. Show that $\left(\frac{10^4}{10^2}\right)^2$ and $\frac{10^8}{10^4}$ have the same value.

IXL

1. Open IXL and start today's skills.
2. Work on paper where the question needs work. The answer box does not show it.
3. Every skill listed is required, to a SmartScore of 67. Due at the start of the next class.

Today's skills — all required

Evaluate expressions using properties of exponents — ZE9

Negative Exponent Law

MA.8.NSO.1.3

I can rewrite a power with a negative exponent using a positive exponent, and evaluate it.

Warm-Up

1. Find the value: $\frac{5^4 \cdot 5^0}{5^2}$
2. Which is greater, $\sqrt{30}$ or 5.5?
3. Solve $-3x = 21$
4. What is the reciprocal of $\frac{3}{4}$? What is the opposite of $\frac{3}{4}$?

Warm-Up

1. Find the value: $\frac{5^4 \cdot 5^0}{5^2}$ **25**
2. Which is greater, $\sqrt{30}$ or 5.5? **5.5**
3. Solve $-3x = 21$ **$x = -7$**
4. What is the reciprocal of $\frac{3}{4}$? What is the opposite of $\frac{3}{4}$?
 $\frac{4}{3}$ and $-\frac{3}{4}$

Notes

I. The pattern keeps going past zero

2^4	2^3	2^2	2^1	2^0	2^{-1}	2^{-2}	2^{-3}
16	8	4	2	1	$\frac{1}{2}$	$\frac{1}{4}$	$\frac{1}{8}$

$$2^{-1} = \frac{1}{2^1}$$

$$2^{-2} = \frac{1}{2^2} = \frac{1}{4}$$

$$2^{-3} = \frac{1}{2^3} = \frac{1}{8}$$

Notes

II. The negative exponent law

$$a^{-n} = \frac{1}{a^n} \quad \frac{1}{a^{-n}} = a^n \quad (a \neq 0)$$

$$5^{-2} = \frac{1}{5^2} = \frac{1}{25} \quad \text{but} \quad -5^2 = -25 \quad \text{and} \quad (-5)^2 = 25$$

A negative exponent means reciprocal. It never means a negative value.

The base keeps its own sign: $(-2)^{-3} = \frac{1}{(-2)^3} = \frac{1}{-8} = -\frac{1}{8}$

Notes

III. The laws still work with negative exponents

$$6^{-2} \cdot 6^5 = \frac{1}{6^2} \cdot 6^5 = \frac{7776}{36} = 216$$

$$\text{or } 6^{-2} \cdot 6^5 = 6^{-2+5} = 6^3 = 216$$

$$\frac{9^{-5}}{9^3} = 9^{-5-3} = 9^{-8} = \frac{1}{9^8}$$

Add, subtract and multiply exponents exactly as before — the exponents are just integers now.

Example 1

$$\frac{2^{-3}}{9^{-2}}$$

Rewrite with positive exponents, then find the value.

Example 1

$$\frac{2^{-3}}{9^{-2}} = \frac{9^2}{2^3}$$

$$\frac{9^2}{2^3} = \frac{81}{8} = 10.125$$

Answer: $81/8 = 10.125$

Your Turn

$$\frac{3^{-2}}{5^{-1}}$$

Rewrite with positive exponents, then find the value.

Your Turn

$$\frac{3^{-2}}{5^{-1}}$$

Rewrite with positive exponents, then find the value.

$$3^{-2} = \frac{1}{3^2} \quad \text{and} \quad \frac{1}{5^{-1}} = 5^1$$

$$\frac{3^{-2}}{5^{-1}} = \frac{5^1}{3^2} = \frac{5}{9}$$

Answer: $\frac{5}{9}$

Example 2

$$\frac{7^0}{7^{-4}}$$

Find the value.

Example 2

$$\frac{7^0}{7^{-4}} = 7^{0 - (-4)} = 7^4$$

$$\frac{7^0}{7^{-4}} = \frac{1}{7^{-4}} = 7^4$$

$$7^4 = 2401$$

Answer: 2,401

Your Turn

$$\frac{(-8)^{-4}}{(-8)^{-6}}$$

Find the value.

Your Turn

$$\frac{(-8)^{-4}}{(-8)^{-6}}$$

Find the value.

$$\frac{(-8)^{-4}}{(-8)^{-6}} = (-8)^{-4 - (-6)} = (-8)^2$$

$$(-8)^2 = 64$$

Answer: 64

Whiteboards · Question 1 of 9

$$8^{-3}$$

Answer it.

Whiteboards • Question 1 of 9

$$8^{-3}$$

$$8^{-3} = \frac{1}{8^3}$$

Answer: $\frac{1}{8^3}$

Whiteboards • Question 2 of 9

A sheet of paper 30 centimeters long is folded in half 7 times.

Each fold makes the top layer half as big.

What fraction of the whole sheet is the top layer after 7 folds?

Answer it.

Whiteboards • Question 2 of 9

A sheet of paper 30 centimeters long is folded in half 7 times.

Each fold makes the top layer half as big.

What fraction of the whole sheet is the top layer after 7 folds?

Each fold halves it: $\left(\frac{1}{2}\right)^7 = \frac{1}{2^7}$

$$2^7 = 128, \text{ so } \frac{1}{2^7} = \frac{1}{128}$$

Answer: $\frac{1}{128}$

Whiteboards • Question 3 of 9

$$\frac{1}{10^{-2}}$$

Answer it.

Whiteboards • Question 3 of 9

$$\frac{1}{10^{-2}}$$

$$\frac{1}{10^{-2}} = 10^2$$

$$10^2 = 100$$

Answer: $10^2 = 100$

Whiteboards · Question 4 of 9

$$(-2)^{-3}$$

Answer it.

Whiteboards • Question 4 of 9

$$(-2)^{-3}$$

$$(-2)^{-3} = \frac{1}{(-2)^3}$$

$$(-2)^3 = -8, \text{ so } \frac{1}{(-2)^3} = -\frac{1}{8}$$

$$\text{Answer: } -\frac{1}{8}$$

Whiteboards · Question 5 of 9

$$5^{-2}$$

Which is the value?

A. 1/10

B. 1/25

C. -25

D. -10

Answer it.

Whiteboards • Question 5 of 9

$$5^{-2}$$

Which is the value?

$$5^{-2} = \frac{1}{5^2} = \frac{1}{25}$$

Answer: B — 1/25

Whiteboards • Question 6 of 9

$$6^{-2} \cdot 6^5$$

Which expression is equivalent?

A. 6^{-10}

B. 36^3

C. 6^{-7}

D. 6^3

Answer it.

Whiteboards • Question 6 of 9

$$6^{-2} \cdot 6^5$$

Which expression is equivalent?

$$6^{-2} \cdot 6^5 = 6^{-2+5} = 6^3$$

Answer: D — 6^3

Whiteboards · Question 7 of 9

$$3^{-4} \cdot 3^4$$

Answer it.

Whiteboards · Question 7 of 9

$$3^{-4} \cdot 3^4$$

$$3^{-4} \cdot 3^4 = 3^{-4+4} = 3^0 = 1$$

Answer: 1

Whiteboards · Question 8 of 9

$$\frac{9^{-5}}{9^3}$$

Answer it.

Whiteboards • Question 8 of 9

$$\frac{9^{-5}}{9^3}$$

$$\frac{9^{-5}}{9^3} = 9^{-5-3} = 9^{-8}$$

$$9^{-8} = \frac{1}{9^8}$$

Answer: $\frac{1}{9^8}$

Whiteboards • Question 9 of 9

Ellen says the value of 5^{-2} is -25 .

Explain why she is wrong, and show the correct value.

Write your answer in sentences.

Whiteboards • Question 9 of 9

Ellen says the value of 5^{-2} is -25 .

Explain why she is wrong, and show the correct value.

A negative exponent means the reciprocal, not the opposite.

$$5^{-2} = \frac{1}{5^2} = \frac{1}{25}$$

Answer: $5^{-2} = 1/5^2 = 1/25$

Independent Set

1. Rewrite with a positive exponent. Do not evaluate. 9^{-2}
2. Find the value. 3^{-4}
3. Find the value. $(-4)^{-2}$
4. Find the value. $\frac{1}{10^{-3}}$
5. Find the value. $5^2 \cdot 5^{-5}$
6. Find the value of n . $2^n = \frac{1}{16}$

IXL

1. Open IXL and start today's skills.
2. Work on paper where the question needs work. The answer box does not show it.
3. Every skill listed is required, to a SmartScore of 67. Due at the start of the next class.

Today's skills — all required

Understanding negative exponents — YBB

Evaluate powers with negative exponents — WGS

Applying Exponent Laws

MA.8.NSO.1.3

I can rewrite a rational base with a negative exponent using a positive exponent, and apply several laws to generate an equivalent expression.

Warm-Up

1. Find the value: 2^{-3}
2. Rewrite with a positive exponent: $\frac{1}{7^{-2}}$
3. Which is larger, 4^3 or 3^4 ?
4. Write the reciprocal of $-\frac{2}{5}$

Warm-Up

1. Find the value: 2^{-3} $\frac{1}{8}$
2. Rewrite with a positive exponent: $\frac{1}{7^{-2}}$ 7^2
3. Which is larger, 4^3 or 3^4 ? 3^4 (**$81 > 64$**)
4. Write the reciprocal of $-\frac{2}{5}$ $-\frac{5}{2}$

Notes

I. The pattern with a fraction base

Expression	Expanded	Value
$(2/5)^2$	$2/5 \cdot 2/5$	$4/25$
$(2/5)^1$	$2/5$	$2/5$
$(2/5)^0$	1	1
$(2/5)^{-1}$	$5/2$	$5/2$
$(2/5)^{-2}$	$5/2 \cdot 5/2$	$25/4$

Notes

II. Flip the base, make the exponent positive

$$\left(\frac{a}{b}\right)^{-n} = \left(\frac{b}{a}\right)^n \quad (a, b \neq 0)$$

$$\left(\frac{2}{5}\right)^{-2} = \left(\frac{5}{2}\right)^2 = \frac{25}{4} \qquad \left(-\frac{2}{3}\right)^{-5} = \left(-\frac{3}{2}\right)^5$$

The sign stays with the fraction. Flip the numbers, not the sign.

A whole number is a fraction over 1: $4^{-3} = \left(\frac{4}{1}\right)^{-3} = \left(\frac{1}{4}\right)^3$

Notes

III. Several laws at once — the plan is still yours

$$\left(\frac{2.5^5}{2.5^3}\right)^{-1} = (2.5^2)^{-1} = 2.5^{-2} = \left(\frac{5}{2}\right)^{-2} = \left(\frac{2}{5}\right)^2 = \frac{4}{25}$$

Any correct order of laws gives the same value. Finish with a positive exponent.

Example 1

$$\left(-\frac{7}{8}\right)^{-2}$$

Rewrite with a positive exponent, then find the value.

Example 1

$$\left(-\frac{7}{8}\right)^{-2} = \left(-\frac{8}{7}\right)^2$$

$$\left(-\frac{8}{7}\right)^2 = \frac{64}{49}$$

Answer: 64/49

Your Turn

$$\left(-\frac{5}{2}\right)^{-3}$$

Rewrite with a positive exponent, then find the value.

Your Turn

$$\left(-\frac{5}{2}\right)^{-3}$$

Rewrite with a positive exponent, then find the value.

Flip the fraction: $\left(-\frac{5}{2}\right)^{-3} = \left(-\frac{2}{5}\right)^3$

$$\left(-\frac{2}{5}\right)^3 = -\frac{8}{125}$$

Answer: $-\frac{8}{125}$

Example 2

$$\left(\frac{5}{8}\right)^{-2} \cdot \left(\frac{5}{8}\right)^4$$

Write an equivalent expression with one base and a positive exponent.

Example 2

$$\left(\frac{5}{8}\right)^{-2} \cdot \left(\frac{5}{8}\right)^4 = \left(\frac{5}{8}\right)^{-2+4} = \left(\frac{5}{8}\right)^2$$

$$\left(\frac{5}{8}\right)^2 = \frac{25}{64}$$

Answer: $(5/8)^2 = 25/64$

Your Turn

$$\left(-\frac{5}{4}\right)^{-7} \cdot \left(-\frac{5}{4}\right)^{-3}$$

Write an equivalent expression with one base and a positive exponent.

Your Turn

$$\left(-\frac{5}{4}\right)^{-7} \cdot \left(-\frac{5}{4}\right)^{-3}$$

Write an equivalent expression with one base and a positive exponent.

Add the exponents, then flip the fraction:

$$\left(-\frac{5}{4}\right)^{-7+(-3)} = \left(-\frac{5}{4}\right)^{-10} = \left(-\frac{4}{5}\right)^{10}$$

Answer: $\left(-\frac{4}{5}\right)^{10}$

Whiteboards • Question 1 of 9

A copier is set to shrink a page to $\frac{2}{3}$ of its size.

A page 6 inches wide is copied 4 times in a row, each copy from the last.

What fraction of the original page's size is the last copy?

Answer it.

Whiteboards • Question 1 of 9

A copier is set to shrink a page to $\frac{2}{3}$ of its size.

A page 6 inches wide is copied 4 times in a row, each copy from the last.

What fraction of the original page's size is the last copy?

Four copies: $\left(\frac{2}{3}\right)^4 = \frac{2^4}{3^4} = \frac{16}{81}$

Answer: $\frac{16}{81}$

Whiteboards • Question 2 of 9

$$\left(\frac{3}{-5}\right)^{-9}$$

Answer it.

Whiteboards • Question 2 of 9

$$\left(\frac{3}{-5}\right)^{-9}$$

Flip the fraction: $\left(\frac{3}{-5}\right)^{-9} = \left(\frac{-5}{3}\right)^9$

Answer: $\left(\frac{-5}{3}\right)^9$

Whiteboards • Question 3 of 9

$$\left(-\frac{8}{9}\right)^{-2}$$

Answer it.

Whiteboards • Question 3 of 9

$$\left(-\frac{8}{9}\right)^{-2}$$

Flip the fraction: $\left(-\frac{8}{9}\right)^{-2} = \left(-\frac{9}{8}\right)^2$

$$\left(-\frac{9}{8}\right)^2 = \frac{81}{64}$$

Answer: $\frac{81}{64}$

Whiteboards • Question 4 of 9

$$\left(\frac{21}{4}\right)^{-5}$$

Answer it.

Whiteboards • Question 4 of 9

$$\left(\frac{21}{4}\right)^{-5}$$

Flip the fraction: $\left(\frac{21}{4}\right)^{-5} = \left(\frac{4}{21}\right)^5$

Answer: $\left(\frac{4}{21}\right)^5$

Whiteboards • Question 5 of 9

$$\left(-\frac{7}{8}\right)^{-3}$$

Which expression is equivalent?

A. $(7/8)^3$

B. $(-8/7)^3$

C. $(-7/8)^3$

D. $(8/7)^3$

Answer it.

Whiteboards • Question 5 of 9

$$\left(-\frac{7}{8}\right)^{-3}$$

Which expression is equivalent?

Flip the fraction. The sign stays: $\left(-\frac{7}{8}\right)^{-3} = \left(-\frac{8}{7}\right)^3$

Answer: B — $(-8/7)^3$

Whiteboards • Question 6 of 9

$$\left(\frac{5}{8}\right)^{-2} \div \left(\frac{5}{8}\right)^{-4}$$

Which expression is equivalent?

A. $(8/5)^2$

B. $(5/8)^{-6}$

C. $(5/8)^2$

D. $(5/8)^8$

Answer it.

Whiteboards • Question 6 of 9

$$\left(\frac{5}{8}\right)^{-2} \div \left(\frac{5}{8}\right)^{-4}$$

Which expression is equivalent?

$$\left(\frac{5}{8}\right)^{-2} \div \left(\frac{5}{8}\right)^{-4} = \left(\frac{5}{8}\right)^{-2 - (-4)} = \left(\frac{5}{8}\right)^2$$

Answer: C — $(5/8)^2$

Whiteboards • Question 7 of 9

$$\left(\frac{2}{5}\right)^{-1} \cdot \left(\frac{5}{2}\right)^1$$

Answer it.

Whiteboards • Question 7 of 9

$$\left(\frac{2}{5}\right)^{-1} \cdot \left(\frac{5}{2}\right)^1$$

$$\left(\frac{2}{5}\right)^{-1} = \left(\frac{5}{2}\right)^1$$

$$\left(\frac{5}{2}\right)^1 \cdot \left(\frac{5}{2}\right)^1 = \left(\frac{5}{2}\right)^2 = \frac{25}{4}$$

Answer: $\frac{25}{4}$

Whiteboards • Question 8 of 9

$$\left[\left(\frac{2}{3} \right)^5 \cdot \left(\frac{2}{3} \right)^{-2} \right]^3$$

Answer it.

Whiteboards • Question 8 of 9

$$\left[\left(\frac{2}{3} \right)^5 \cdot \left(\frac{2}{3} \right)^{-2} \right]^3$$

Inside: $\left(\frac{2}{3} \right)^5 \cdot \left(\frac{2}{3} \right)^{-2} = \left(\frac{2}{3} \right)^{5 + (-2)} = \left(\frac{2}{3} \right)^3$

$$\left(\left(\frac{2}{3} \right)^3 \right)^3 = \left(\frac{2}{3} \right)^{3 \cdot 3} = \left(\frac{2}{3} \right)^9$$

Answer: $\left(\frac{2}{3} \right)^9$

Whiteboards • Question 9 of 9

Marco says $\left(-\frac{2}{3}\right)^{-5}$ can be rewritten as $\left(\frac{2}{3}\right)^5$

because 'two negatives make a positive.' Is he right?

Write your answer in sentences.

Whiteboards • Question 9 of 9

Marco says $\left(-\frac{2}{3}\right)^{-5}$ can be rewritten as $\left(\frac{2}{3}\right)^5$

because 'two negatives make a positive.' Is he right?

The negative exponent flips the fraction. The sign stays.

$$\left(-\frac{2}{3}\right)^{-5} = \left(-\frac{3}{2}\right)^5$$

Answer: No — $(-2/3)^{-5} = (-3/2)^5$

Independent Set

1. Rewrite with a positive exponent. $\left(\frac{3}{4}\right)^{-2}$
2. Find the value. $\left(\frac{2}{5}\right)^{-2}$
3. Rewrite with a positive exponent. $\left(-\frac{5}{6}\right)^{-3}$
4. Find the value. $\left(\frac{1}{2}\right)^{-4}$
5. Find the value. $\left(\frac{3}{2}\right)^{-1} \cdot \left(\frac{3}{2}\right)^3$
6. Find the value of n . $\left(\frac{4}{7}\right)^n = \left(\frac{7}{4}\right)^3$

IXL

1. Open IXL and start today's skills.
2. Work on paper where the question needs work. The answer box does not show it.
3. Every skill listed is required, to a SmartScore of 67. Due at the start of the next class.

Today's skills — all required

Identify equivalent expressions involving exponents I — VLM

Evaluating and Equivalent Expressions

MA.8.NSO.1.3

I can evaluate an expression with integer exponents using the laws, and decide whether two expressions with different bases are equivalent.

Warm-Up

1. Rewrite with a positive exponent, then evaluate: $\left(\frac{2}{3}\right)^{-2}$
2. Find the value: $\frac{1}{5^{-2}}$
3. Find the value: $\frac{10^6}{10^4}$
4. Write 64 as a power of 2, of 4, and of 8.

Warm-Up

1. Rewrite with a positive exponent, then evaluate: $\left(\frac{2}{3}\right)^{-2}$
 $\left(\frac{3}{2}\right)^2 = \frac{9}{4}$
2. Find the value: $\frac{1}{5^{-2}}$ **25**
3. Find the value: $\frac{10^6}{10^4}$ **100**
4. Write 64 as a power of 2, of 4, and of 8. **2^6 , 4^3 , 8^2**

Notes

I. Two students, two plans, one value

Melanie: $\frac{5^{-2} \cdot 5^5}{5} = \frac{5^3}{5} = 5^2 = 25$

Adonis: $\frac{5^{-2} \cdot 5^5}{5} = \frac{\frac{1}{25} \cdot 3125}{5} = \frac{125}{5} = 25$

Apply the laws first, evaluate last. The value is the same; the arithmetic is smaller.

Notes

II. Every law, one expression

$$\frac{(6^2)^{-3}}{6^{-5}} = \frac{6^{-6}}{6^{-5}} = 6^{-6 - (-5)} = 6^{-1} = \frac{1}{6}$$

Find the step that broke:

$$\left(\frac{1}{5}\right)^4 \cdot \left(\frac{1}{5}\right)^{-2} = \left(\frac{1}{5}\right)^{-8} \quad \text{— wrong: multiplied; should be } \left(\frac{1}{5}\right)^2$$

$$\left(\frac{3}{2}\right)^{-3} = \frac{2^{-3}}{3^{-3}} \quad \text{— wrong: flipped, kept the minus; should be } \left(\frac{2}{3}\right)^3$$

Notes

III. Different bases, same value — rewrite to a common base

Base	Square	Cube	Fourth
2	$2^2 = 4$	$2^3 = 8$	$2^4 = 16$
3	$3^2 = 9$	$3^3 = 27$	$3^4 = 81$
5	$5^2 = 25$	$5^3 = 125$	$5^4 = 625$

$$8^5 = (2^3)^5 = 2^{15} \qquad 9^4 = (3^2)^4 = 3^8$$

$$2^3 \cdot 4^3 = 2^3 \cdot (2^2)^3 = 2^3 \cdot 2^6 = 2^9$$

Notes

IV. Testing equivalence: Xavier's claim

$$\left(\frac{1}{4}\right)^2 \cdot 2^{-3} = 2^{-4} \cdot 2^{-3} = 2^{-7}$$

$$\left(\frac{1}{2}\right)^4 \cdot 4^{-1} = 2^{-4} \cdot 2^{-2} = 2^{-6}$$

Same base, then compare exponents.

$2^{-7} \neq 2^{-6}$ — so Xavier is wrong.

Example 1

$$\left(\frac{3}{4}\right)^{-2} \cdot 3$$

Find the value.

Example 1

$$\left(\frac{3}{4}\right)^{-2} \cdot 3 = \left(\frac{4}{3}\right)^2 \cdot 3$$

$$\frac{16}{9} \cdot 3 = \frac{16}{3}$$

Answer: 16/3

Your Turn

$$\left(\frac{2}{5}\right)^{-2} \cdot 2$$

Find the value.

Your Turn

$$\left(\frac{2}{5}\right)^{-2} \cdot 2$$

Find the value.

Flip the fraction: $\left(\frac{2}{5}\right)^{-2} = \left(\frac{5}{2}\right)^2 = \frac{25}{4}$

$$\frac{25}{4} \cdot 2 = \frac{25}{2}$$

Answer: $\frac{25}{2}$

Example 2

Are $3^2 \cdot 6^3$ and $\frac{(3 \cdot 2)^5}{2^2}$ equivalent?

Rewrite both with the bases 3 and 2, then compare.

Example 2

$$3^2 \cdot 6^3 = 3^2 \cdot 3^3 \cdot 2^3 = 3^5 \cdot 2^3$$

$$\frac{(3 \cdot 2)^5}{2^2} = \frac{3^5 \cdot 2^5}{2^2} = 3^5 \cdot 2^3$$

$$3^5 \cdot 2^3 = 3^5 \cdot 2^3$$

Answer: Yes — both are $3^5 \cdot 2^3 = 1944$

Your Turn

Are $2^2 \cdot 10^3$ and $\frac{(2 \cdot 5)^5}{5^2}$ equivalent?

Your Turn

Are $2^2 \cdot 10^3$ and $\frac{(2 \cdot 5)^5}{5^2}$ equivalent?

$$2^2 \cdot 10^3 = 2^2 \cdot 2^3 \cdot 5^3 = 2^5 \cdot 5^3$$

$$\frac{(2 \cdot 5)^5}{5^2} = \frac{2^5 \cdot 5^5}{5^2} = 2^5 \cdot 5^3$$

$$2^5 \cdot 5^3 = 32 \cdot 125 = 4,000$$

Answer: Yes — $2^5 \cdot 5^3 = 4000$

Whiteboards • Question 1 of 9

$$\frac{(-7)^5}{(-7)^0 \cdot (-7)^2}$$

Answer it.

Whiteboards • Question 1 of 9

$$\frac{(-7)^5}{(-7)^0 \cdot (-7)^2}$$

Bottom: $(-7)^0 \cdot (-7)^2 = (-7)^2$

$$\frac{(-7)^5}{(-7)^2} = (-7)^{5-2} = (-7)^3 = -343$$

Answer: -343

Whiteboards • Question 2 of 9

$$\frac{\left(\frac{3}{4}\right)^{-3} \cdot \left(\frac{3}{4}\right)^0}{\left(\frac{3}{4}\right)^{-2}}$$

Answer it.

Whiteboards • Question 2 of 9

$$\frac{\left(\frac{3}{4}\right)^{-3} \cdot \left(\frac{3}{4}\right)^0}{\left(\frac{3}{4}\right)^{-2}}$$

Top: $\left(\frac{3}{4}\right)^{-3} \cdot \left(\frac{3}{4}\right)^0 = \left(\frac{3}{4}\right)^{-3}$

$$\left(\frac{3}{4}\right)^{-3 - (-2)} = \left(\frac{3}{4}\right)^{-1} = \frac{4}{3}$$

Answer: $\frac{4}{3}$

Whiteboards • Question 3 of 9

$$\frac{(2^{-1})^3 \cdot \left(\frac{1}{2}\right)^{-4}}{2 \cdot 2^2}$$

Answer it.

Whiteboards • Question 3 of 9

$$\frac{(2^{-1})^3 \cdot \left(\frac{1}{2}\right)^{-4}}{2 \cdot 2^2}$$

Top: $(2^{-1})^3 = 2^{-3}$ and $\left(\frac{1}{2}\right)^{-4} = 2^4$, so $2^{-3} \cdot 2^4 = 2^1$

Bottom: $2 \cdot 2^2 = 2^3$

$$\frac{2^1}{2^3} = 2^{1-3} = 2^{-2} = \frac{1}{4}$$

Answer: $\frac{1}{4}$

Whiteboards • Question 4 of 9

One flash drive holds 2^{15} megabytes.

Another holds 8^5 megabytes. The first one costs 45 dollars.

Which holds more — or is it the same? Show why.

Answer it.

Whiteboards • Question 4 of 9

One flash drive holds 2^{15} megabytes.

Another holds 8^5 megabytes. The first one costs 45 dollars.

Which holds more — or is it the same? Show why.

$$8 = 2^3$$

$$8^5 = (2^3)^5 = 2^{3 \cdot 5} = 2^{15}$$

Answer: The same: $8^5 = (2^3)^5 = 2^{15}$

Whiteboards • Question 5 of 9

$$9^4$$

Which expression is equivalent?

A. 3^{16}

B. 3^4

C. 3^8

D. 3^6

Answer it.

Whiteboards • Question 5 of 9

$$9^4$$

Which expression is equivalent?

$$9 = 3^2$$

$$9^4 = (3^2)^4 = 3^{2 \cdot 4} = 3^8$$

Answer: C — 3^8

Whiteboards • Question 6 of 9

$$\left(\frac{1}{4}\right)^2 \cdot 2^{-3}$$

Which expression is equivalent?

A. 2^{-7}

B. 2^7

C. 2^{-5}

D. 2^{-12}

Answer it.

Whiteboards • Question 6 of 9

$$\left(\frac{1}{4}\right)^2 \cdot 2^{-3}$$

Which expression is equivalent?

$$\left(\frac{1}{4}\right)^2 = (2^{-2})^2 = 2^{-4}$$

$$2^{-4} \cdot 2^{-3} = 2^{-4 + (-3)} = 2^{-7}$$

Answer: A — 2^{-7}

Whiteboards · Question 7 of 9

$$2^3 \cdot 4^3$$

Answer it.

Whiteboards • Question 7 of 9

$$2^3 \cdot 4^3$$

$$4^3 = (2^2)^3 = 2^6$$

$$2^3 \cdot 2^6 = 2^{3+6} = 2^9$$

Answer: 2^9

Whiteboards · Question 8 of 9

$$4^7 \cdot 4^{-2}$$

Answer it.

Whiteboards • Question 8 of 9

$$4^7 \cdot 4^{-2}$$

$$4^7 \cdot 4^{-2} = 4^{7+(-2)} = 4^5$$

$$4^5 = (2^2)^5 = 2^{10}$$

Answer: 2^{10}

Whiteboards • Question 9 of 9

Xavier says $\left(\frac{1}{4}\right)^2 \cdot 2^{-3}$ and $\left(\frac{1}{2}\right)^4 \cdot 4^{-1}$ are equivalent because both can be written as 2^{-7} . Do you agree?

Write your answer in sentences.

Whiteboards • Question 9 of 9

Xavier says $\left(\frac{1}{4}\right)^2 \cdot 2^{-3}$ and $\left(\frac{1}{2}\right)^4 \cdot 4^{-1}$ are equivalent

because both can be written as 2^{-7} . Do you agree?

$$\text{First: } \left(\frac{1}{4}\right)^2 \cdot 2^{-3} = 2^{-4} \cdot 2^{-3} = 2^{-7}$$

$$\text{Second: } \left(\frac{1}{2}\right)^4 \cdot 4^{-1} = 2^{-4} \cdot 2^{-2} = 2^{-6}$$

Answer: No — the second is 2^{-6} , not 2^{-7}

Independent Set

1. Find the value. $\frac{(-5)^4}{(-5)^0 \cdot (-5)^2}$
2. Find the value. $\left(\frac{2}{3}\right)^{-2} \cdot \left(\frac{2}{3}\right)^3$
3. Write as a power of 2. 16^3
4. Write as a power of 3. $27^2 \cdot 9$
5. Are 4^{-2} and 2^{-4} equivalent? Show why.
6. Find the value. $(3^{-1})^2 \cdot 3^5$

IXL

1. Open IXL and start today's skills.
2. Work on paper where the question needs work. The answer box does not show it.
3. Every skill listed is required, to a SmartScore of 67. Due at the start of the next class.

Today's skills — all required

Evaluate expressions using properties of exponents — 6T7

Identify equivalent expressions involving exponents II — QDM

Exponent Laws with Variable Bases

MA.8.AR.1.1

I can multiply and divide monomials, and raise a monomial to a power, by applying the laws of exponents to variable bases.

Warm-Up

1. Find the value: $(2^3)^2 \cdot 2^{-4}$
2. Write as one power: $\frac{6^7}{6^5}$
3. $\sqrt{50}$ is between which two whole numbers?
4. Write with an exponent: $Y \cdot Y \cdot Y \cdot Y \cdot Y$

Warm-Up

1. Find the value: $(2^3)^2 \cdot 2^{-4}$ **4**
2. Write as one power: $\frac{6^7}{6^5}$ **$6^2 = 36$**
3. $\sqrt{50}$ is between which two whole numbers? **7 and 8**
4. Write with an exponent: $y \cdot y \cdot y \cdot y \cdot y$ **y^5**

Notes

I. Product of powers — the base is a letter now

$$x^2 \cdot x^3 = (x \cdot x)(x \cdot x \cdot x) = x^5 \qquad x^m \cdot x^n = x^{m+n}$$

$$3x^2 \cdot 5x^4 = (3 \cdot 5)(x^2 \cdot x^4) = 15x^6$$

Multiply the coefficients. Add the exponents of like bases.

Different variables stay separate: $x^2y \cdot x^3y^4 = x^5y^5$

Notes

II. Power of a power, power of a product

$$(x^3)^4 = x^3 \cdot x^3 \cdot x^3 \cdot x^3 = x^{12}$$

$$(x^m)^n = x^{mn}$$

$$(2y^2)^3 = 2^3 \cdot (y^2)^3 = 8y^6$$

$$(15xy^2)^3 = 3,375x^3y^6$$

Power of a power: multiply the exponents. Power of a product: raise every factor.

Notes

III. Quotient of powers — dividing monomials

$$\frac{x^7}{x^2} = \frac{x \cdot x \cdot x \cdot x \cdot x \cdot x \cdot x}{x \cdot x} = x^5$$

$$\frac{x^m}{x^n} = x^{m-n}$$

$$\frac{12a^5}{4a^2} = 3a^3$$

$$\frac{m^6n^4}{m^2n^4} = m^4n^0 = m^4$$

Divide the coefficients. Subtract the exponents of like bases.

Example 1

$$(4x^3y)(-2x^2y^5)$$

Write an equivalent expression as a single monomial.

Example 1

$$\begin{aligned}(4x^3y)(-2x^2y^5) &= (4)(-2) \cdot x^3x^2 \cdot y^1y^5 \\ &= -8 \cdot x^5 \cdot y^6 = -8x^5y^6\end{aligned}$$

Answer: $-8x^5y^6$

Your Turn

$$(5a^2b^4)(3ab^2)$$

Write an equivalent expression as a single monomial.

Your Turn

$$(5a^2b^4)(3ab^2)$$

Write an equivalent expression as a single monomial.

Numbers: $5 \cdot 3 = 15$

The a's: $a^2 \cdot a^1 = a^{2+1} = a^3$

The b's: $b^4 \cdot b^2 = b^{4+2} = b^6$

Answer: $15a^3b^6$

Example 2

$$\frac{(3m^2n)^3}{9m^4n}$$

Write an equivalent expression as a single monomial.

Example 2

$$(3m^2n)^3 = 27m^6n^3$$

$$\frac{27m^6n^3}{9m^4n} = 3 \cdot m^{6-4} \cdot n^{3-1} = 3m^2n^2$$

Answer: $3m^2n^2$

Your Turn

$$\frac{(2p^3)^4}{8p^5}$$

Write an equivalent expression as a single monomial.

Your Turn

$$\frac{(2p^3)^4}{8p^5}$$

Write an equivalent expression as a single monomial.

Top first: $(2p^3)^4 = 2^4 \cdot p^{3 \cdot 4} = 16p^{12}$

Numbers: $16 \div 8 = 2$

The p's: $p^{12} \div p^5 = p^{12-5} = p^7$

Answer: $2p^7$

Whiteboards · Question 1 of 9

$$x^5 \cdot x^8$$

Answer it.

Whiteboards • Question 1 of 9

$$x^5 \cdot x^8$$

$$x^5 \cdot x^8 = x^{5+8} = x^{13}$$

Answer: x^{13}

Whiteboards • Question 2 of 9

$$(y^4)^3$$

Answer it.

Whiteboards • Question 2 of 9

$$(y^4)^3$$

$$(y^4)^3 = y^{4 \cdot 3} = y^{12}$$

Answer: y^{12}

Whiteboards • Question 3 of 9

A cube has edges $2x^3$ centimeters long.

Its surface area is $24x^6$ square centimeters.

What is the volume of the cube, in cubic centimeters?

Answer it.

Whiteboards • Question 3 of 9

A cube has edges $2x^3$ centimeters long.

Its surface area is $24x^6$ square centimeters.

What is the volume of the cube, in cubic centimeters?

Volume = edge $\times$ edge $\times$ edge: $(2x^3)^3$

$$(2x^3)^3 = 2^3 \cdot x^{3 \cdot 3} = 8x^9$$

Answer: $8x^9$

Whiteboards • Question 4 of 9

$$(3a^2b)(4ab^5)$$

Answer it.

Whiteboards · Question 4 of 9

$$(3a^2b)(4ab^5)$$

Numbers: $3 \cdot 4 = 12$

The a's: $a^2 \cdot a^1 = a^{2+1} = a^3$

The b's: $b^1 \cdot b^5 = b^{1+5} = b^6$

Answer: $12a^3b^6$

Whiteboards • Question 5 of 9

$$(5m^3)^2$$

Which expression is equivalent?

A. $25m^6$

B. $10m^6$

C. $25m^5$

D. $5m^6$

Answer it.

Whiteboards • Question 5 of 9

$$(5m^3)^2$$

Which expression is equivalent?

$$(5m^3)^2 = 5^2 \cdot m^{3 \cdot 2} = 25m^6$$

Answer: A — $25m^6$

Whiteboards • Question 6 of 9

$$\frac{12x^8}{3x^2}$$

Which expression is equivalent?

A. $4x^{10}$

B. $4x^6$

C. $9x^6$

D. $4x^4$

Answer it.

Whiteboards • Question 6 of 9

$$\frac{12x^8}{3x^2}$$

Which expression is equivalent?

Numbers: $12 \div 3 = 4$ The x's: $x^8 \div x^2 = x^{8-2} = x^6$

Answer: B — $4x^6$

Whiteboards • Question 7 of 9

$$\frac{m^6 n^4}{m^2 n^4}$$

Answer it.

Whiteboards • Question 7 of 9

$$\frac{m^6 n^4}{m^2 n^4}$$

The m's: $m^6 \div m^2 = m^{6-2} = m^4$

The n's: $n^4 \div n^4 = n^{4-4} = n^0 = 1$

Answer: m^4

Whiteboards • Question 8 of 9

Find the value of n :

$$x^3 \cdot x^n = x^{11}$$

Answer it.

Whiteboards · Question 8 of 9

Find the value of n :

$$x^3 \cdot x^n = x^{11}$$

Same base, so the exponents add: $3 + n = 11$

$$n = 11 - 3 = 8$$

Answer: $n = 8$

Whiteboards • Question 9 of 9

Rachel writes $(15xy^2)^3$ as $15 \cdot 15 \cdot 15 \cdot x \cdot x \cdot x \cdot y \cdot y \cdot y \cdot y \cdot y \cdot y$.

Justina writes it as $3,375x^3y^6$. Who is correct? Explain.

Write your answer in sentences.

Whiteboards • Question 9 of 9

Rachel writes $(15xy^2)^3$ as $15 \cdot 15 \cdot 15 \cdot x \cdot x \cdot x \cdot y \cdot y \cdot y \cdot y \cdot y \cdot y$.

Justina writes it as $3,375x^3y^6$. Who is correct? Explain.

$$15 \cdot 15 \cdot 15 = 3,375$$

$$x \cdot x \cdot x = x^3$$

$$y \cdot y \cdot y \cdot y \cdot y \cdot y = y^6$$

Answer: Both are correct; $15^3 = 3,375$, $x \cdot x \cdot x = x^3$, and six y 's make y^6 .

Independent Set

1. Write as one power. $x^6 \cdot x^9$
2. Write as a single monomial. $(a^5)^4$
3. Write as a single monomial. $(3y^2)^3$
4. Write as a single monomial. $(2m^4)(5m^3)$
5. Write as a single monomial. $\frac{20x^9}{4x^3}$
6. Find the value of n . $y^4 \cdot y^n = y^{10}$

IXL

1. Open IXL and start today's skills.
2. Work on paper where the question needs work. The answer box does not show it.
3. Every skill listed is required, to a SmartScore of 67. Due at the start of the next class.

Today's skills — all required

Multiply powers: variable bases — TR9

Powers of a power: variable bases — 2CU

Divide monomials with positive exponents — C8L

Negative Exponents with Variable Bases

MA.8.AR.1.1

I can rewrite an algebraic expression with negative exponents using only positive exponents, and apply several laws of exponents to a monomial expression.

Warm-Up

1. Write as a single monomial: $(3x^2)^3$
2. Find the value: 5^{-2}
3. Which is greater, $\sqrt{80}$ or 9?
4. What is the reciprocal of $\frac{2}{3}$?

Warm-Up

1. Write as a single monomial: $(3x^2)^3$ **$27x^6$**
2. Find the value: 5^{-2} **$1/25$**
3. Which is greater, $\sqrt{80}$ or 9? **9**
4. What is the reciprocal of $\frac{2}{3}$? **$\frac{3}{2}$**

Notes

I. The pattern, with a letter

x^4	x^3	x^2	x^1	x^0	x^{-1}	x^{-2}	x^{-3}
$x \cdot x \cdot x \cdot x$	$x \cdot x \cdot x$	$x \cdot x$	x	1	$1/x$	$1/x^2$	$1/x^3$

$$x^{-n} = \frac{1}{x^n} \quad \frac{1}{x^{-n}} = x^n \quad (x \neq 0)$$

Notes

II. Moving factors across the fraction bar

$$3x^{-2} = \frac{3}{x^2}$$

$$y^{-3}z^{-4} = \frac{1}{y^3z^4}$$

$$\frac{a^2}{b^{-3}} = a^2b^3$$

$$\frac{5x^{-4}y^2}{z^{-1}} = \frac{5y^2z}{x^4}$$

Only the factor with the negative exponent moves. Its exponent turns positive.

Notes

III. The laws with negative exponents

$$x^4 \cdot x^{-6} = x^{4+(-6)} = x^{-2} = \frac{1}{x^2} \qquad \frac{m^3}{m^7} = m^{3-7} = m^{-4} = \frac{1}{m^4}$$

$$(2y^{-3})^2 = 2^2 \cdot y^{-6} = \frac{4}{y^6}$$

Apply the law. Then rewrite so that every exponent is positive.

Example 1

$$(2x^{-3}y^2)(6x^5y^{-4})$$

Write an equivalent expression with positive exponents only.

Example 1

$$(2x^{-3}y^2)(6x^5y^{-4}) = 12 \cdot x^{-3+5} \cdot y^{2+(-4)} \\ = 12x^2y^{-2} = \frac{12x^2}{y^2}$$

Answer: $12x^2/y^2$

Your Turn

$$(4a^{-2}b)(a^5b^{-3})$$

Write an equivalent expression with positive exponents only.

Your Turn

$$(4a^{-2}b)(a^5b^{-3})$$

Write an equivalent expression with positive exponents only.

The a's: $a^{-2} \cdot a^5 = a^{-2+5} = a^3$

The b's: $b^1 \cdot b^{-3} = b^{1+(-3)} = b^{-2} = \frac{1}{b^2}$

$$4 \cdot a^3 \cdot \frac{1}{b^2} = \frac{4a^3}{b^2}$$

Answer: $\frac{4a^3}{b^2}$

Example 2

$$\left(\frac{a^3}{4b^{-7}}\right)^5$$

Write an equivalent expression with positive exponents only.

Example 2

$$\left(\frac{a^3}{4b^{-7}}\right)^5 = \frac{a^{15}}{4^5 b^{-35}}$$
$$= \frac{a^{15} b^{35}}{1,024}$$

Answer: $a^{15}b^{35}/1,024$

Your Turn

$$\left(\frac{x^2}{3y^{-2}}\right)^3$$

Write an equivalent expression with positive exponents only.

Your Turn

$$\left(\frac{x^2}{3y^{-2}} \right)^3$$

Write an equivalent expression with positive exponents only.

Cube every part, then move the y up:

$$\left(\frac{x^2}{3y^{-2}} \right)^3 = \frac{x^6}{27y^{-6}} = \frac{x^6 y^6}{27}$$

Answer: $\frac{x^6 y^6}{27}$

Whiteboards · Question 1 of 9

$$x^{-4}$$

Answer it.

Whiteboards • Question 1 of 9

$$x^{-4}$$
$$x^{-4} = \frac{1}{x^4}$$

Answer: $\frac{1}{x^4}$

Whiteboards • Question 2 of 9

$$\frac{1}{y^{-5}}$$

Answer it.

Whiteboards • Question 2 of 9

$$\frac{1}{y^{-5}}$$

$$\frac{1}{y^{-5}} = y^5$$

Answer: y^5

Whiteboards • Question 3 of 9

The brightness of a lamp from x meters away is $3x^{-2}$ units.

The lamp is 15 centimeters tall.

How bright is the lamp from 2 meters away?

Answer it.

Whiteboards • Question 3 of 9

The brightness of a lamp from x meters away is $3x^{-2}$ units.

The lamp is 15 centimeters tall.

How bright is the lamp from 2 meters away?

$$3x^{-2} = \frac{3}{x^2}$$

$$\text{When } x = 2: \frac{3}{2^2} = \frac{3}{4}$$

$$\text{Answer: } \frac{3}{4}$$

Whiteboards • Question 4 of 9

$$y^{-3}z^{-4}$$

Answer it.

Whiteboards • Question 4 of 9

$$y^{-3}z^{-4}$$

$$y^{-3}z^{-4} = \frac{1}{y^3} \cdot \frac{1}{z^4} = \frac{1}{y^3z^4}$$

Answer: $\frac{1}{y^3z^4}$

Whiteboards • Question 5 of 9

$$x^4 \cdot x^{-6}$$

Which expression is equivalent?

A. $-x^2$

B. $1/x^{24}$

C. $1/x^2$

D. x^{10}

Answer it.

Whiteboards • Question 5 of 9

$$x^4 \cdot x^{-6}$$

Which expression is equivalent?

$$x^4 \cdot x^{-6} = x^{4 + (-6)} = x^{-2}$$

$$x^{-2} = \frac{1}{x^2}$$

Answer: C — $1/x^2$

Whiteboards • Question 6 of 9

$$(2x^{-3})^2$$

Which expression is equivalent?

A. $4/x$

B. $1/(4x^6)$

C. $2/x^6$

D. $4/x^6$

Answer it.

Whiteboards • Question 6 of 9

$$(2x^{-3})^2$$

Which expression is equivalent?

$$(2x^{-3})^2 = 2^2 \cdot x^{-3 \cdot 2} = 4x^{-6}$$

$$4x^{-6} = \frac{4}{x^6}$$

Answer: D — $4/x^6$

Whiteboards • Question 7 of 9

$$\frac{m^3}{m^7}$$

Answer it.

Whiteboards • Question 7 of 9

$$\frac{m^3}{m^7}$$

$$\frac{m^3}{m^7} = m^{3-7} = m^{-4}$$

$$m^{-4} = \frac{1}{m^4}$$

Answer: $\frac{1}{m^4}$

Whiteboards • Question 8 of 9

Find the value of n :

$$x^{-5} \cdot x^n = x^2$$

Answer it.

Whiteboards · Question 8 of 9

Find the value of n :

$$x^{-5} \cdot x^n = x^2$$

Same base, so the exponents add: $-5 + n = 2$

$$n = 2 + 5 = 7$$

Answer: $n = 7$

Whiteboards • Question 9 of 9

How are x^3 and x^{-3} related?

Explain, and give the value of each when $x = 2$.

Write your answer in sentences.

Whiteboards • Question 9 of 9

How are x^3 and x^{-3} related?

Explain, and give the value of each when $x = 2$.

$x^{-3} = \frac{1}{x^3}$, the reciprocal of x^3

When $x = 2$: $2^3 = 8$ and $2^{-3} = \frac{1}{8}$

Answer: x^{-3} is the reciprocal of x^3 . When $x = 2$: $x^3 = 8$ and $x^{-3} = 1/8$.

Independent Set

1. Rewrite using only positive exponents. m^{-6}
2. Rewrite using only positive exponents. $5y^{-3}$
3. Rewrite using only positive exponents. $\frac{1}{x^{-7}}$
4. Simplify. Use positive exponents only. $x^2 \cdot x^{-9}$
5. Simplify. Use positive exponents only. $(3m^{-2})^3$
6. Find the value of n . $y^{-3} \cdot y^n = y^4$

IXL

1. Open IXL and start today's skills.
2. Work on paper where the question needs work. The answer box does not show it.
3. Every skill listed is required, to a SmartScore of 67. Due at the start of the next class.

Today's skills — all required

Variables with negative exponents — 2NL

Multiply and divide powers: variable bases — 85P

Writing Large Numbers in Scientific Notation

MA.8.NSO.1.4

I can write a large number in scientific notation, read one back into standard form, and find how many times larger one number is than another.

Warm-Up

1. Rewrite with a positive exponent: $4x^{-3}$
2. Find the value: $(10^2)^3$
3. Find the value: $\sqrt{121}$
4. In 4,500,000, which place is the 4 in?

Warm-Up

1. Rewrite with a positive exponent: $4x^{-3}$ **$4/x^3$**
2. Find the value: $(10^2)^3$ **$10^6 = 1,000,000$**
3. Find the value: $\sqrt{121}$ **11**
4. In 4,500,000, which place is the 4 in? **millions**

Notes

I. Powers of ten name the places

10^1	10^2	10^3	10^4	10^5	10^6
10	100	1,000	10,000	100,000	1,000,000

The exponent counts the zeros — and names the place of the 1.

Notes

II. Scientific notation

$$a \times 10^n \text{ with } 1 \leq a < 10$$

$$2,000,000 = 2 \times 10^6 \qquad 26,000,000 = 2.6 \times 10^7$$

The coefficient is at least 1 and less than 10: one nonzero digit before the decimal point.

The exponent is how many places the decimal point moves: 26,000,000. → 2.6, seven places, so 10^7 .

Notes

III. How many times larger?

$$\frac{6.4 \times 10^7}{3.2 \times 10^7} = 2$$

$$\frac{2 \times 10^5}{2 \times 10^2} = 10^3 = 1,000$$

$$\frac{8.4 \times 10^6}{2.1 \times 10^4} = 4 \times 10^2 = 400$$

Divide the coefficients. Divide the powers of 10. Multiply the two results.

On a screen, 8.4E6 means 8.4×10^6 .

Example 1

In 2018, funding for education in Florida totaled 30,100,000,000 dollars.

Express this value in scientific notation.

Example 1

$$30,100,000,000 = 3.01 \times 10^{10}$$

$$3.01 \times 10,000,000,000 = 30,100,000,000$$

Answer: 3.01×10^{10}

Your Turn

A painting sold for 110,500,000 dollars. Write this value in scientific notation.

Your Turn

A painting sold for 110,500,000 dollars. Write this value in scientific notation.

Move the point 8 places left: 1.105

$$1.105 \times 10^8 = 110,500,000$$

Answer: 1.105×10^8

Example 2

How many times larger is 8.1×10^8 than 5.4×10^7 ?

Example 2

$$\frac{8.1}{5.4} = 1.5$$

$$\frac{10^8}{10^7} = 10^1 = 10$$

$$1.5 \times 10 = 15$$

Answer: 15 times larger

Your Turn

Everglades alligators: 2.0×10^5

Florida alligators: 1.5×10^6

About how many times larger is the Florida number?

Your Turn

Everglades alligators: 2.0×10^5

Florida alligators: 1.5×10^6

About how many times larger is the Florida number?

Divide the numbers: $1.5 \div 2.0 = 0.75$

Divide the powers: $10^6 \div 10^5 = 10^1$

$0.75 \times 10^1 = 7.5$

Answer: about 7.5 times larger

Whiteboards • Question 1 of 9

Write 4,000,000 in scientific notation.

Answer it.

Whiteboards · Question 1 of 9

Write 4,000,000 in scientific notation.

The 4 is in the millions place: $1,000,000 = 10^6$

$$4 \times 10^6 = 4,000,000$$

Answer: 4×10^6

Whiteboards • Question 2 of 9

Write 7.2×10^4 in standard form.

Answer it.

Whiteboards • Question 2 of 9

Write 7.2×10^4 in standard form.

$$10^4 = 10,000$$

$$7.2 \times 10,000 = 72,000$$

Answer: 72,000

Whiteboards • Question 3 of 9

A stadium's renovation used 356,000 bolts.

The stadium seats 71,000 people.

Write the number of bolts in scientific notation.

Answer it.

Whiteboards • Question 3 of 9

A stadium's renovation used 356,000 bolts.

The stadium seats 71,000 people.

Write the number of bolts in scientific notation.

Move the point 5 places left: 3.56

$$3.56 \times 10^5 = 356,000$$

Answer: 3.56×10^5

Whiteboards · Question 4 of 9

Which is greater, 9.5×10^5 or 1.2×10^6 ?

Answer it.

Whiteboards · Question 4 of 9

Which is greater, 9.5×10^5 or 1.2×10^6 ?

$$9.5 \times 10^5 = 950,000$$

$$1.2 \times 10^6 = 1,200,000$$

Answer: 1.2×10^6

Whiteboards • Question 5 of 9

Which is 52,000,000 written in scientific notation?

A. 5.2×10^8

B. 5.2×10^7

C. 52×10^6

D. 5.2×10^6

Answer it.

Whiteboards • Question 5 of 9

Which is 52,000,000 written in scientific notation?

Move the point 7 places left: 5.2

$$5.2 \times 10^7 = 52,000,000$$

Answer: B — 5.2×10^7

Whiteboards • Question 6 of 9

How many times larger is 6×10^8 than 6×10^5 ?

A. 1.6

B. 3

C. 1,000

D. 300

Answer it.

Whiteboards • Question 6 of 9

How many times larger is 6×10^8 than 6×10^5 ?

$$6 \div 6 = 1$$

$$10^8 \div 10^5 = 10^{8-5} = 10^3 = 1,000$$

Answer: C — 1,000

Whiteboards • Question 7 of 9

How many times larger is 8.4×10^6 than 2.1×10^4 ?

Answer it.

Whiteboards • Question 7 of 9

How many times larger is 8.4×10^6 than 2.1×10^4 ?

Divide the numbers: $8.4 \div 2.1 = 4$

Divide the powers: $10^6 \div 10^4 = 10^2$

$$4 \times 10^2 = 400$$

Answer: 400

Whiteboards • Question 8 of 9

A computer displays a number as $3.5\text{E}9$.

Write the number in standard form.

Answer it.

Whiteboards • Question 8 of 9

A computer displays a number as 3.5E9.

Write the number in standard form.

3.5E9 means 3.5×10^9

$3.5 \times 10^9 = 3,500,000,000$

Answer: 3,500,000,000

Whiteboards • Question 9 of 9

Marcus writes 2,000,000 as 20×10^5 .

Is his expression equal to 2,000,000? Is it scientific notation? Explain.

Write your answer in sentences.

Whiteboards • Question 9 of 9

Marcus writes 2,000,000 as 20×10^5 .

Is his expression equal to 2,000,000? Is it scientific notation? Explain.

$20 \times 10^5 = 2,000,000$, so it is equal.

The first number must be at least 1 and less than 10. 20 is not.

$2 \times 10^6 = 2,000,000$

Answer: Equal, yes. Scientific notation, no: 20 is not less than 10. 2×10^6 .

Independent Set

1. Write 8,200,000 in scientific notation.
2. Write 6.4×10^5 in standard form.
3. Write 915,000,000 in scientific notation.
4. Which is greater, 4.7×10^7 or 9.3×10^6 ?
5. How many times larger is 9×10^7 than 3×10^4 ?
6. A computer displays a number as 2.6E7. Write it in standard form.

IXL

1. Open IXL and start today's skills.
2. Work on paper where the question needs work. The answer box does not show it.
3. Every skill listed is required, to a SmartScore of 67. Due at the start of the next class.

Today's skills — all required

Convert between standard and scientific notation: large numbers — 2AL

Compare large numbers — WEZ

Powers of ten — LJQ

Writing Small Numbers in Scientific Notation

MA.8.NSO.1.4

I can write very small numbers in scientific notation, read them back, and find how many times larger or smaller one number is than another.

Warm-Up

1. Write 4,300,000 in scientific notation.
2. Write 10^{-3} as a fraction.
3. $\sqrt{95}$ is between which two whole numbers?
4. Write 0.007 as a fraction.

Warm-Up

1. Write 4,300,000 in scientific notation. 4.3×10^6
2. Write 10^{-3} as a fraction. $\frac{1}{1,000}$
3. $\sqrt{95}$ is between which two whole numbers? **9 and 10**
4. Write 0.007 as a fraction. $\frac{7}{1,000}$

Notes

I. Negative powers of ten name the decimal places

10^0	10^{-1}	10^{-2}	10^{-3}	10^{-4}	10^{-5}
1	0.1	0.01	0.001	0.0001	0.00001

$$10^{-n} = \frac{1}{10^n} \qquad 10^{-3} = \frac{1}{1,000} = 0.001$$

A negative exponent puts the 1 in that decimal place: 10^{-4} is a 1 in the fourth place.

Notes

II. Small numbers in scientific notation

$$0.000007 = 7 \times 10^{-6}$$

$$0.0000672 = 6.72 \times 10^{-5}$$

$$3.4 \times 10^{-9} = 0.0000000034$$

The coefficient is still at least 1 and less than 10. The exponent is negative for a number less than 1.

The exponent is how many places the point moves: $0.0000672 \rightarrow 6.72$ is five places right, so 10^{-5} .

Notes

III. How many times larger — or smaller?

$$\frac{7.2 \times 10^{-4}}{1.8 \times 10^{-7}} = 4 \times 10^3 = 4,000$$

$$\frac{3 \times 10^8}{3 \times 10^{-4}} = 10^{12}$$

Divide the coefficients. Divide the powers of 10 — subtract the exponents. Multiply the results.

Which is bigger, 10^{-5} or 10^{-8} ? The one with the larger exponent: $-5 > -8$, so 10^{-5} .

Example 1

The density of oxygen is 1,332 millionths of a gram per cubic centimeter.

Write this value as a decimal, then in scientific notation.

Example 1

$$\frac{1,332}{1,000,000} = 0.001332$$

$$0.001332 = 1.332 \times 10^{-3}$$

Answer: $0.001332 = 1.332 \times 10^{-3}$

Your Turn

A human hair is about 67 hundred-thousandths of an inch thick.

Write this value in scientific notation.

Your Turn

A human hair is about 67 hundred-thousandths of an inch thick.

Write this value in scientific notation.

67 hundred-thousandths = 0.00067

Move the point 4 places right: 6.7

$$6.7 \times 10^{-4} = 0.00067$$

Answer: 6.7×10^{-4}

Example 2

How many times larger is 7.2×10^{-4} than 1.8×10^{-7} ?

Example 2

$$\frac{7.2}{1.8} = 4$$

$$\frac{10^{-4}}{10^{-7}} = 10^{-4 - (-7)} = 10^3$$

$$4 \times 10^3 = 4,000$$

Answer: 4,000 times larger

Your Turn

How many times larger is 5.2×10^{-3} than 2.6×10^{-8} ?

Your Turn

How many times larger is 5.2×10^{-3} than 2.6×10^{-8} ?

Divide the numbers: $5.2 \div 2.6 = 2$

Divide the powers: $10^{-3} \div 10^{-8} = 10^{-3 - (-8)} = 10^5$

$2 \times 10^5 = 200,000$

Answer: 200,000 times larger

Whiteboards · Question 1 of 9

$$10^{-4}$$

Answer it.

Whiteboards • Question 1 of 9

$$10^{-4}$$
$$10^{-4} = \frac{1}{10^4} = \frac{1}{10,000} = 0.0001$$

Answer: 0.0001

Whiteboards • Question 2 of 9

Write 0.003 in scientific notation.

Answer it.

Whiteboards · Question 2 of 9

Write 0.003 in scientific notation.

The 3 is in the thousandths place: $0.001 = 10^{-3}$

$$3 \times 10^{-3} = 0.003$$

Answer: 3×10^{-3}

Whiteboards • Question 3 of 9

Write 2.5×10^{-4} in standard form.

Answer it.

Whiteboards • Question 3 of 9

Write 2.5×10^{-4} in standard form.

$$10^{-4} = 0.0001$$

$$2.5 \times 0.0001 = 0.00025$$

Answer: 0.00025

Whiteboards • Question 4 of 9

A machine's cutting tolerance is 0.0000988 inch.

The part it cuts is 4 inches long.

Write the tolerance in scientific notation.

Answer it.

Whiteboards · Question 4 of 9

A machine's cutting tolerance is 0.0000988 inch.

The part it cuts is 4 inches long.

Write the tolerance in scientific notation.

Move the point 5 places right: 9.88

$$9.88 \times 10^{-5} = 0.0000988$$

Answer: 9.88×10^{-5}

Whiteboards • Question 5 of 9

Which is 0.00081 written in scientific notation?

A. 8.1×10^{-3}

B. 8.1×10^4

C. 81×10^{-5}

D. 8.1×10^{-4}

Answer it.

Whiteboards • Question 5 of 9

Which is 0.00081 written in scientific notation?

Move the point 4 places right: 8.1

$$8.1 \times 10^{-4} = 0.00081$$

Answer: D — 8.1×10^{-4}

Whiteboards • Question 6 of 9

How many times larger is 6×10^{-3} than 2×10^{-5} ?

A. 300

B. 30

C. 3

D. 0.03

Answer it.

Whiteboards • Question 6 of 9

How many times larger is 6×10^{-3} than 2×10^{-5} ?

Divide the numbers: $6 \div 2 = 3$

Divide the powers: $10^{-3} \div 10^{-5} = 10^{-3 - (-5)} = 10^2$

$$3 \times 10^2 = 300$$

Answer: A — 300

Whiteboards • Question 7 of 9

Which is greater, 3.9×10^{-5} or 7.4×10^{-8} ?

Answer it.

Whiteboards • Question 7 of 9

Which is greater, 3.9×10^{-5} or 7.4×10^{-8} ?

$$3.9 \times 10^{-5} = 0.000039$$

$$7.4 \times 10^{-8} = 0.000000074$$

Answer: 3.9×10^{-5}

Whiteboards • Question 8 of 9

A computer displays a number as $4.2\text{E}-6$.

Write the number in standard form.

Answer it.

Whiteboards • Question 8 of 9

A computer displays a number as $4.2\text{E}-6$.

Write the number in standard form.

$4.2\text{E}-6$ means 4.2×10^{-6}

$4.2 \times 10^{-6} = 0.0000042$

Answer: 0.0000042

Whiteboards • Question 9 of 9

Fishing line A is 2×10^{-2} inch thick.

Fishing line B is 6×10^{-2} inch thick.

Which is thicker, and how many times? Name a thickness between them.

Write your answer in sentences.

Whiteboards · Question 9 of 9

Fishing line A is 2×10^{-2} inch thick.

Fishing line B is 6×10^{-2} inch thick.

Which is thicker, and how many times? Name a thickness between them.

Same power of 10, so compare 2 and 6.

$$6 \div 2 = 3$$

Between them: 4×10^{-2}

Answer: B; 3 times; e.g. 4×10^{-2} in.

Independent Set

1. Write 0.00047 in scientific notation.
2. Write 5.2×10^{-3} in standard form.
3. Write 0.0000061 in scientific notation.
4. Which is greater, 2.4×10^{-6} or 8.1×10^{-9} ?
5. How many times larger is 8×10^{-2} than 4×10^{-5} ?
6. A computer displays a number as 3.7E-5. Write it in standard form.

IXL

1. Open IXL and start today's skills.
2. Work on paper where the question needs work. The answer box does not show it.
3. Every skill listed is required, to a SmartScore of 67. Due at the start of the next class.

Today's skills — all required

Convert between standard and scientific notation: small numbers — 85M

Compare small numbers — KDR

Powers of ten with negative exponents — SKD